

Aniket Ghosh

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Education

M.S. in Artificial Intelligence, ML Concentration – Northeastern University, Khoury College – GPA: 4.0/4.0 Aug 2025 – May 2027
Coursework: CS 7180 Actionable Interpretable Methods (PhD seminar, Prof. Byron Wallace) • CS 5180 Reinforcement Learning • CS 5100 Foundations of AI • CS 5800 Algorithms • IE 7374 Machine Learning Operations (MLOps).

B.Tech in CS & Business Systems – Institute of Engineering & Management, Kolkata – GPA: 4.0/4.0 Jul 2020 – Jun 2024
Ranked **2nd of 180+** • **Director's Award** • *Key Courses:* Neural Networks, NLP, Data Structures, OS.

Research, Publications & Patents

Function Vectors as a Model-Diffing Tool: 17 Heads Repair a Bad Fine-Tune *Mech Interp* • [LessWrong](#) 2026 • [GitHub](#)

• Patched function vectors **across models** rather than contexts: transplanting **17 attention heads** took a misaligned medical LoRA from **3 to 27 of 36** safe responses against a clean-model ceiling of 30, **ARC flat at 100%**, beating crosscoder diffing (1), few-shot (23) and steering (10). Repairs misalignment but cannot induce it; replicated on **Llama-3.1-8B**.

Linear Probes Tell You Where Quantization Will Hurt *LLM Compression* • [LessWrong](#) 2026 • [GitHub](#)

• Trained per-layer linear probes (NER, POS, chunking) to localize token-level signal, then allocated precision by layer: held **99–100% accuracy at a 5-bit average** where uniform quantization collapsed to **16–41%** on three unseen datasets.

Intelligent Traffic Sign Detection Using YOLOv8 with Hybrid CNN Filtering *First Author*

AISC 2024 • Springer Proceedings in Information and Communication Technologies (in press, Oct 2026)

Lightweight Hybrid DNN-GNN for Network Intrusion Detection with Adaptive Late Fusion *Co-author*

Published – IEEE Xplore • **Patent** Filed by SRM Institute of Science and Technology

Technical Skills

Interpretability: nnsight, NDIF, PyTorch hooks, DAS / Boundless DAS / MDAS, causal patching, linear probes, crosscoders, SAEs

Training: PyTorch, Hugging Face Transformers, accelerate, distributed multi-GPU training, SLURM, LoRA/QLoRA, quantization

GenAI & Agents: RAG, LangChain, LangGraph, CrewAI, OpenAI Agents SDK, MCP, ChromaDB, Claude Code, Cursor

MLOps: Docker, Kubernetes, AWS, GCP, CI/CD, MLflow, FastAPI, Terraform

Core Domains: Mechanistic Interpretability, AI Safety, Reinforcement Learning, CV, NLP, Multi-Agent Systems, Medical Imaging

Languages: Python, C/C++, SQL, Bash

Experience

Research Mentee, Interpretability – *SPAR (Supervised Program for AI Research)* Sep 2026 – Dec 2026

• Selected for **A Diagnostic Panel of Ground-Truth Probes for Language Models** with Mohamed Amine Merzouk (Mila) and Adam Oberman (McGill): probes restricted to targets with **exact, computable ground truth** — remaining response length, end-of-sequence hazard — scored on **calibration** rather than noisy human labels.

AI/ML Engineer – *Varosync, Inc., San Francisco, CA* (remote from Boston) May 2026 – Aug 2026

• **Sole ML engineer** at an early-stage drug-discovery company, reporting to the CEO/CTO. Built the stack from scratch: a **biomedical knowledge graph** over millions of molecules, then featurization, **vector indexing**, retrieval and the live user query path.

• Trained **7–8B parameter** molecular embedding models on an **800M-molecule** corpus across a **multi-GPU cluster**, owning the whole loop from data curation and distributed training through checkpointing and evaluation.

• Moved published research into production: the **per-layer probing method** from the quantization work above shipped into those models to decide where capacity goes across layers. Paper in preparation.

Teaching Co-Lead, ML Lab – *AI & Data Ethics Program (AIDE), Northeastern Ethics Institute* May 2026 – Aug 2026

• Designed and taught a **12-session ML lab** for a residential cohort of philosophy PhD students in this **Sloan Foundation-funded** program on fairness, differential privacy, transformers, epistemic simulations and mechanistic interpretability.

• Taught **sparse autoencoders, the logit lens, and linear probing** hands-on using **nnsight** and **NDIF**.

AI Researcher – *CMATER Lab, Jadavpur University, Kolkata* Jun 2023 – May 2025

• Built an unsupervised pipeline for histopathological cell segmentation that reached **85%+ accuracy** with **30% fewer false positives** than a Mask R-CNN baseline, enabling annotation-free pathology screening.

• Designed the preprocessing stage: **color quantization** plus **DBSCAN** to separate nuclei that intensity thresholding merges together.

• Worked two years under Dr. Kaushiki Roy and Prof. Debotosh Bhattacharjee; mentored juniors in deep learning for medical imaging.

Selected Projects

Sufficient Cause Disambiguation Mechanistic Interpretability • CS 7180 (PhD seminar) • [GitHub](#)

• Proved **probing accuracy and causal relevance are distinct properties** on Llama-3-8B: separator tokens hit **1.000 LDA accuracy** across all 32 layers yet drive a **0% flip rate** under causal patching.

• Built the pipeline (LDA, gradient attribution, causal patching): **448× compression at 99.6% accuracy, 100% flip rate** at $\alpha=2$.

• Audited **LLM resume screening for demographic bias**: name-swapped resumes triggered a **95.8% flip rate** and accounted for **62%** of the model's highest-uncertainty Fit decisions on otherwise identical content.

Biomedical Knowledge Graph Link Prediction Healthcare AI • CS 5100 • [GitHub](#)

• Engineered graph features (PageRank, structural metrics) on the **BioRED** biomedical corpus for a Random Forest link predictor at **0.94 ROC-AUC**, with **5-hop** path reasoning for relationship explainability.

• Ran a classical ML vs. **TransE** embedding benchmark (**0.94** vs. **0.61** ROC-AUC), showing hand-crafted graph features dominate learned representations on low-resource biomedical corpora, a practical finding for data-scarce settings.

Autonomous Multi-Agent Trading Simulation [GitHub](#)

• Designed and deployed a multi-agent trading simulation on the **OpenAI Agents SDK**: **4 autonomous AI traders** coordinated across **6 MCP servers** with **44 distinct tools**, consuming live market data via **Polygon.io** for autonomous portfolio management.

Achievements & Service

BlueDot Impact Technical AI Safety — certified July 2026; the function-vectors work above was the Project Sprint output • **Graduate**

TA, NLP, Northeastern Khoury — Ran PyTorch and deep learning labs for **40+** graduate students • **Volunteer Educator, The Hope**

Foundation — 7 years teaching math and CS to underprivileged students, Kolkata • **Graduate Student Senator**, Northeastern GSG.